

Numeraire-Based Distributional Sensitivity Engine for European Options under Stochastic Interest Rates*

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Abstract

Distributional Sensitivity Engine (DSE) [16] organizes European option prices and sensitivities for equity diffusions with deterministic term structures by rewriting the in-the-money event through a Doleans–Dade exponential. The price is expressed in the two-weight form $V_t = w_1 a_1 + w_2 a_2$, where (a_1, a_2) are $\mathcal{F}_t$ -measurable primitives and (w_1, w_2) are conditional digital quantities under two measures linked by a Girsanov tilt. A limitation of DSE is the assumption that discounting is $\mathcal{F}_t$ -measurable.

This paper presents an extension of that representation to European options under stochastic interest rates. A central step normalizes the discount factor by the bond price $B(t, T) = \mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T}]$. This normalization yields an $\mathcal{F}_t$ -measurable exercise boundary for a modified Doleans–Dade log variable $\tilde{E}_{t,T}$. The spot leg remains under the Doleans tilt measure $\mathbb{Q}^+$, while the strike leg is written through a digital weight under the T -forward measure, or more generally under a suitable numeraire measure. The resulting representation retains the basic engine structure. In particular, it retains a two-weight price representation and a four-input contraction form for first derivatives. The probabilistic core is therefore reduced to conditional distribution blocks under $\mathbb{Q}^+$ and $\mathbb{Q}^T$. Dirac-delta representations are then used to construct Monte Carlo estimators for CDF, PDF, and derivative blocks, and a mollified delta yields estimators with the standard bias–variance scaling associated with kernel smoothing.

Keywords: Greeks, European options, stochastic interest rates, stochastic discounting, change of numeraire, forward measures, Dirac delta, Monte Carlo

JEL: G13, C02, C63

1 Introduction

BSM Sensitivity Engine (BSE) [15] introduces a reusable chain-rule template for first and mixed second derivatives of the Black–Scholes–Merton price [1, 2]. The representation takes the form of a dot product between common weights and derivatives of a small collection of primitive inputs.

*This paper continues *A Distributional Sensitivity Engine (DSE)* by David Hongkai Shen (February 2026), which follows *BSM Sensitivity Engine (BSE)* by David Hongkai Shen (January 2026).

It also isolates an explicit correction channel that accounts for the motion of the weights. *Distributional Sensitivity Engine* (DSE) extends the same organization beyond closed-form BSM [16] by expressing prices through conditional distribution blocks of a Doleans–Dade log variable under two measures linked by a Girsanov tilt. In DSE, discounting is exogenous and $\mathcal{F}_t$ -measurable, so the discount factor $\text{DF}_{t,T}$ can be taken outside conditional expectations.

The present paper studies an extension of that framework to stochastic interest rates. Discounting is therefore allowed to depend on a stochastic short rate. A central step replaces $\text{DF}_{t,T}$ with the normalized factor $\text{DF}_{t,T}/B(t,T)$, where $B(t,T) = \mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T}]$. This normalization yields an $\mathcal{F}_t$ -measurable exercise boundary for a modified Doleans–Dade log variable $\tilde{E}_{t,T}$. The strike leg then admits a representation under a suitable numeraire measure. Standard change-of-numeraire arguments support the derivation, and the analysis focuses on the resulting engine representation together with the associated differentiation logic under stochastic discounting.

Three structural features are retained. The option value retains the two-weight form $V_t = w_1 a_1 + w_2 a_2$. The first-derivative formula retains the four-input contraction structure of DSE. The probabilistic blocks move from $(\mathbb{Q}, \mathbb{Q}^+)$ to $(\mathbb{Q}^T, \mathbb{Q}^+)$, while a general numeraire measure may replace $\mathbb{Q}^T$ in the strike leg. The term-structure component remains modular. HJM provides one specification, while LMM and SMM correspond to alternative numeraire choices.

Dividend convention. The main text assumes a deterministic dividend yield $q(\cdot)$, with either a constant level or a deterministic term structure. The stochastic-discounting analysis in the main text is presented under deterministic dividend yield assumptions.

2 Setup: stochastic discounting under an arbitrage-free term structure

Fix $0 \leq t < T$ and let $D \in \{+1, -1\}$ denote call/put sign. Consider the unified European payoff

$$f_T = (D(S_T - K))^+. \quad (1)$$

Let $(\mathcal{F}_u)_{u \geq 0}$ support a risk-neutral measure $\mathbb{Q}$ associated with the money-market numeraire. Assume a deterministic dividend yield $q(u)$ and define the money market account and discount factor

$$M_u := \exp\left(\int_0^u r_s ds\right), \quad \text{DF}_{t,T} := \frac{M_t}{M_T} = \exp\left(-\int_t^T r_s ds\right). \quad (2)$$

Risk-neutral valuation then takes the standard form [10]

$$V_t = \mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T} f_T]. \quad (3)$$

Arbitrage-free term structure in diffusion form. Let W^r be a d_r -dimensional $\mathbb{Q}$ -Brownian motion driving the term-structure factors. Assume that, for each maturity $U \geq u$, the zero-coupon bond price $B(u, U) > 0$ exists and admits $\mathbb{Q}$ -dynamics of the form [4, 6]

$$\frac{dB(u, U)}{B(u, U)} = r_u du - \Sigma(u, U)^\top dW_u^r, \quad (4)$$

where $\Sigma(u, U) \in \mathbb{R}^{d_r}$ is the model-implied bond-volatility vector. In particular,

$$B(t, T) = \mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T}], \quad (5)$$

so the bond price remains the $\mathcal{F}_t$ -measurable discounting primitive in the engine, as in DSE.

T -forward measure. The T -forward measure $\mathbb{Q}^T$ is defined by

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} \Big|_{\mathcal{F}_T} = \frac{\text{DF}_{t,T}}{B(t, T)}. \quad (6)$$

Equivalently, define the *density process* on $[t, T]$ by

$$L_u^T := \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \Big|_{\mathcal{F}_u} = \frac{\text{DF}_{t,u} B(u, T)}{B(t, T)}, \quad u \in [t, T], \quad (7)$$

Here $\text{DF}_{t,u} := \exp(-\int_t^u r_s ds) = M_t/M_u$ for $u \in [t, T]$, so that $L_t^T = 1$ and $L_T^T = \text{DF}_{t,T}/B(t, T)$. Denote the terminal value by $L^T := L_T^T = \text{DF}_{t,T}/B(t, T)$.

Under the bond dynamics (4), the density process admits the exponential-martingale form

$$L_u^T = \mathcal{E}\left(-\int_t^u \Sigma(s, T)^\top dW_s^r\right) = \exp\left(-\int_t^u \Sigma(s, T)^\top dW_s^r - \frac{1}{2} \int_t^u \|\Sigma(s, T)\|^2 ds\right), \quad u \in [t, T], \quad (8)$$

under standard square-integrability (e.g. Novikov) conditions [8, 9]. In particular, evaluation of (8) at $u = T$ yields $L_T^T = \text{DF}_{t,T}/B(t, T)$, that is (6).

Equity dynamics. Let W^S be a d_S -dimensional Brownian motion, possibly correlated with W^r , and let $W = (W^S, W^r)$ denote the joint Brownian driver under $\mathbb{Q}$. For instance, one may take $d\langle W^{S,i}, W^{r,j} \rangle_u = \rho_{ij}(u) du$ with a bounded predictable correlation process (ρ_{ij}) . Assume an equity diffusion of the generic form

$$dS_u = S_u (r_u - q(u)) du + S_u \sigma_S(u, \cdot)^\top dW_u^S, \quad (9)$$

where σ_S may depend on state variables, including S_u and rate factors. The *Numeraire-Based Distributional Sensitivity Engine* (NDSE) representation requires only that the joint model defines $\text{DF}_{t,T}$ and S_T on a common probability space so that the expectations in (3) are well-defined.

3 Event rewriting with a Doleans–Dade log variable

As in *Distributional Sensitivity Engine* (DSE), write the terminal spot through a Doleans–Dade exponential. From (9),

$$S_T = S_t \exp\left(\int_t^T (r_u - q(u)) du\right) \mathcal{E}\left(\int_t^T \sigma_S(u, \cdot)^\top dW_u^S\right). \quad (10)$$

The definitions $\text{DF}_{t,T} = \exp(-\int_t^T r_u du)$ and $Q_{t,T} := \exp(\int_t^T q(u) du)$ give

$$S_T = \frac{S_t}{\text{DF}_{t,T}} \cdot \frac{1}{Q_{t,T}} \cdot \mathcal{E}\left(\int_t^T \sigma_S(u, \cdot)^\top dW_u^S\right). \quad (11)$$

Define the Doleans–Dade log variable, as in DSE ,

$$E_{t,T} := -\ln \mathcal{E} \left(\int_t^T \sigma_S(u, \cdot)^\top dW_u^S \right). \quad (12)$$

In DSE , deterministic term structures make the exercise boundary m $\mathcal{F}_t$ -measurable. With stochastic rates, the boundary inherits randomness through $\text{DF}_{t,T}$. To restore an $\mathcal{F}_t$ -measurable boundary while keeping the same threshold-event structure, define the *numeraire-adjusted* log variable

$$\tilde{E}_{t,T} := E_{t,T} + \ln \left(\frac{\text{DF}_{t,T}}{B(t,T)} \right), \quad \tilde{m} := \ln \left(\frac{S_t}{K} \right) - \ln Q_{t,T} - \ln B(t,T) = \ln \left(\frac{S_t}{K Q_{t,T} B(t,T)} \right). \quad (13)$$

Since $Q_{t,T}$ is deterministic under the assumption on q and $B(t,T)$ is $\mathcal{F}_t$ -measurable, the boundary $\tilde{m}$ in (13) is $\mathcal{F}_t$ -measurable.

Combining (11) and (13) yields the identity

$$\tilde{E}_{t,T} - \tilde{m} = \ln \left(\frac{K}{S_T} \right). \quad (14)$$

HJM explicit form. Under the HJM specification, the bond-normalization term in (13) is explicitly a Brownian functional. From (8),

$$\ln \left(\frac{\text{DF}_{t,T}}{B(t,T)} \right) = - \int_t^T \Sigma(u, T)^\top dW_u^r - \frac{1}{2} \int_t^T \|\Sigma(u, T)\|^2 du. \quad (15)$$

Hence $\tilde{E}_{t,T} = E_{t,T} + \ln(\text{DF}_{t,T}/B(t,T))$ is the sum of an equity-exponential log term and an HJM bond-exponential log term. This decomposition is convenient for simulation and for likelihood-ratio reweighting.

Equations (11)–(13) yield the in-the-money event

$$\mathbf{1}_{\{DS_T > DK\}} = \mathbf{1}_{\{D\tilde{E}_{t,T} < D\tilde{m}\}}. \quad (16)$$

This is the stochastic-discounting analogue of the event rewriting in DSE . The bond-price normalization turns the boundary into an $\mathcal{F}_t$ -measurable scalar.

When the conditional law of $\tilde{E}_{t,T}$ is continuous, the choice between strict and weak inequalities is immaterial since $\mathbb{P}_t(\tilde{E}_{t,T} = \tilde{m}) = 0$ under any equivalent measure.

4 Two-weight price under stochastic discounting

Starting from (3), expand the payoff as a signed difference:

$$V_t = D \mathbb{E}_t^{\mathbb{Q}} [\text{DF}_{t,T} S_T \mathbf{1}_{\{DS_T > DK\}}] - D K \mathbb{E}_t^{\mathbb{Q}} [\text{DF}_{t,T} \mathbf{1}_{\{DS_T > DK\}}]. \quad (17)$$

4.1 Spot leg: Girsanov tilt as in DSE

Equation (11) implies that the rate integral cancels in $\text{DF}_{t,T} S_T$:

$$\text{DF}_{t,T} S_T = \frac{S_t}{Q_{t,T}} \mathcal{E} \left(\int_t^T \sigma_S(u, \cdot)^\top dW_u^S \right) = \frac{S_t}{Q_{t,T}} e^{-E_{t,T}}. \quad (18)$$

Assume the exponential-martingale condition needed for the Doleans exponential (e.g., Novikov), define $\mathbb{Q}^+$ via

$$\frac{d\mathbb{Q}^+}{d\mathbb{Q}} \Big|_{\mathcal{F}_T} = \mathcal{E} \left(\int_t^T \sigma_S(u, \cdot)^\top dW_u^S \right) = e^{-E_{t,T}}. \quad (19)$$

Correlation remark. If the equity and rate drivers are correlated, the change of measure (19) is defined on the joint filtration and shifts the full Brownian driver accordingly; in particular, the rate factors may acquire an additional drift under $\mathbb{Q}^+$. The spot-leg identity (20) remains a pure change-of-measure statement and does not require writing the drift under $\mathbb{Q}^+$ explicitly.

Then the first term in (17) becomes

$$D \mathbb{E}_t^{\mathbb{Q}} [\text{DF}_{t,T} S_T \mathbf{1}_{\{DS_T > DK\}}] = \underbrace{\frac{S_t}{Q_{t,T}}}_{=:a_1} \cdot \underbrace{D \mathbb{E}_t^{\mathbb{Q}^+} [\mathbf{1}_{\{D\tilde{E}_{t,T} < D\tilde{m}\}}]}_{=:w_1}. \quad (20)$$

Relative to *Distributional Sensitivity Engine* (DSE), the prefactor a_1 is unchanged. The only modification is that the event is written in terms of $\tilde{E}_{t,T}$ and the $\mathcal{F}_t$ -measurable boundary $\tilde{m}$.

4.2 Strike leg: Bayes change-of-numeraire identity

The second term in (17) involves stochastic discounting. Apply the standard Bayes change-of-numeraire identity [5, 7] with the bond price (5):

$$\mathbb{E}_t^{\mathbb{Q}} [\text{DF}_{t,T} X_T] = B(t, T) \mathbb{E}_t^{\mathbb{Q}^T} [X_T], \quad \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \Big|_{\mathcal{F}_T} = \frac{\text{DF}_{t,T}}{B(t, T)}. \quad (21)$$

Correlation remark. When equity and rate factors are correlated, the change from $\mathbb{Q}$ to the T -forward measure $\mathbb{Q}^T$ changes the joint drift of all factors. In Monte Carlo, it is typically simplest to simulate under $\mathbb{Q}$ and reweight by $L^T = d\mathbb{Q}^T/d\mathbb{Q}|_{\mathcal{F}_T}$ (Section 6). This choice avoids explicit drift rewriting under $\mathbb{Q}^T$.

Choosing $X_T = \mathbf{1}_{\{D\tilde{E}_{t,T} < D\tilde{m}\}}$ yields

$$-DK \mathbb{E}_t^{\mathbb{Q}} [\text{DF}_{t,T} \mathbf{1}_{\{DS_T > DK\}}] = \underbrace{-KB(t, T)}_{=:a_2} \cdot \underbrace{D \mathbb{E}_t^{\mathbb{Q}^T} [\mathbf{1}_{\{D\tilde{E}_{t,T} < D\tilde{m}\}}]}_{=:w_2}. \quad (22)$$

4.3 Two-weight form

Combining (20) and (22) yields the same structural representation as DSE :

$$\begin{aligned} V_t &= w_1 a_1 + w_2 a_2, \\ a_1 &= \frac{S_t}{Q_{t,T}}, & a_2 &= -KB(t, T), \\ w_1 &= D \mathbb{P}_t^{\mathbb{Q}^+} (D\tilde{E}_{t,T} < D\tilde{m}), & w_2 &= D \mathbb{P}_t^{\mathbb{Q}^T} (D\tilde{E}_{t,T} < D\tilde{m}). \end{aligned} \quad (23)$$

4.4 General numeraire version

For a strictly positive numeraire process with terminal value N_T , the same argument yields

$$\mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T} X_T] = \widetilde{\text{DF}}_{t,T}^{(N)} \mathbb{E}_t^{\mathbb{Q}^N} \left[\frac{X_T}{N_T} \right], \quad \widetilde{\text{DF}}_{t,T}^{(N)} := \mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T} N_T], \quad \frac{d\mathbb{Q}^N}{d\mathbb{Q}} \Big|_{\mathcal{F}_T} = \frac{\text{DF}_{t,T} N_T}{\mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T} N_T]}. \quad (24)$$

Here $\widetilde{\text{DF}}_{t,T}^{(N)}$ is the time- t price of the terminal numeraire payoff under $\mathbb{Q}$, so (24) is the standard change-of-numeraire identity written in terminal-payoff form. If N is a traded numeraire process, then $\widetilde{\text{DF}}_{t,T}^{(N)} = N_t$. Setting $N_T \equiv 1$ gives $\widetilde{\text{DF}}_{t,T}^{(N)} = \mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T}] = B(t, T)$, and the induced measure $\mathbb{Q}^N$ coincides with the T -forward measure $\mathbb{Q}^T$ because $\frac{d\mathbb{Q}^N}{d\mathbb{Q}} \Big|_{\mathcal{F}_T} = \frac{\text{DF}_{t,T}}{B(t, T)}$. Thus (24) reduces to (21).

Remark 4.1. Arbitrage-free term structures with HJM input specification.

The *Numerator-Based Distributional Sensitivity Engine* (NDSE) derivation in Sections 2–4 is not tied to a particular short-rate model. It requires an arbitrage-free term structure in the form of a strictly positive bond price

$$B(t, T) = \mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T}], \quad (25)$$

so that the T -forward measure $\mathbb{Q}^T$ is well-defined by

$$\mathbb{E}_t^{\mathbb{Q}^T}[\text{DF}_{t,T} X_T] = B(t, T) \mathbb{E}_t^{\mathbb{Q}^T}[X_T], \quad \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \Big|_{\mathcal{F}_T} = \frac{\text{DF}_{t,T}}{B(t, T)}. \quad (26)$$

Consequently, any arbitrage-free term-structure model can enter the engine through the inputs $(B(t, T), \mathbb{Q}^T)$ and, when needed, through the joint law with the equity factors.

A generic Heath–Jarrow–Morton (HJM) specification fits this framework directly. Let the instantaneous forward rate curve be $f(t, T) := -\partial_T \ln B(t, T)$ and, under $\mathbb{Q}$, postulate the HJM dynamics

$$df(t, T) = \alpha(t, T) dt + \sigma_f(t, T) \cdot dW_t^r, \quad t \leq T, \quad (27)$$

with the HJM drift restriction, which ensures no arbitrage under the money-market numeraire,

$$\alpha(t, T) = \sigma_f(t, T) \cdot \int_t^T \sigma_f(t, u) du, \quad (28)$$

where $\cdot$ denotes the multi-factor inner product.

Then

$$B(t, T) = \exp \left(- \int_t^T f(t, u) du \right), \quad \Sigma(t, T) := \int_t^T \sigma_f(t, u) du, \quad (29)$$

and the bond has diffusion

$$\frac{dB(t, T)}{B(t, T)} = r_t dt - \Sigma(t, T)^\top dW_t^r. \quad (30)$$

In particular, the T -forward likelihood ratio admits the exponential-martingale representation

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} \Big|_{\mathcal{F}_T} = \frac{\text{DF}_{t,T}}{B(t, T)} = \mathcal{E} \left(- \int_t^T \Sigma(u, T)^\top dW_u^r \right), \quad (31)$$

so the strike-leg digital weight in (22) is determined once $\Sigma(\cdot, T)$ is specified.

From this viewpoint, HJM serves as an input specification for the NDSE term-structure component. Choosing σ_f and hence Σ determines $B(t, T)$ and the forward-measure change (31). Common short-rate models correspond to induced HJM volatility structures. For example, one-factor Hull–White implies a *deterministic* bond volatility $\Sigma(t, T)$, and therefore a deterministic σ_f , whereas CIR implies an *state-dependent* bond volatility of the form

$$\frac{dB(t, T)}{B(t, T)} = r_t dt - \Sigma^{\text{CIR}}(t, T) dW_t^r, \quad \Sigma^{\text{CIR}}(t, T) = \sigma_r \beta^{\text{CIR}}(t, T) \sqrt{r_t}, \quad (32)$$

Here $\beta^{\text{CIR}}(t, T)$ denotes the standard affine coefficient in the CIR bond price

$$B(t, T) = A^{\text{CIR}}(t, T) \exp\{-\beta^{\text{CIR}}(t, T)r_t\}.$$

This corresponds to an HJM model with stochastic adapted forward-rate volatility. Within the NDSE setting, HJM may therefore be viewed as a special case. It offers one general way to generate the bond input $B(t, T)$ and the associated forward-measure reweighting required by the engine.

Remark 4.2. Discrete-tenor forward measures and the connection to the Libor market model (LMM). Fix a tenor grid $0 = T_0 < T_1 < \dots < T_n$ with accrual factors $\alpha_i := T_{i+1} - T_i$ and define the forward Libor

$$L_i(t) := L(t; T_i, T_{i+1}) = \frac{1}{\alpha_i} \left(\frac{B(t, T_i)}{B(t, T_{i+1})} - 1 \right).$$

The LMM models $(L_i)_{i=0}^{n-1}$ directly and is naturally expressed under the forward measure $\mathbb{Q}^{T_{i+1}}$ associated with the numeraire $B(\cdot, T_{i+1})$. This is the specialization of (21) with $T = T_{i+1}$, so the strike-leg digital blocks in (23) can be evaluated under the same forward measures used in LMM. The present paper is model-agnostic. LMM offers one possible rate module for specifying $(B, \mathbb{Q}^T)$ on a discrete tenor.

Remark 4.3. Annuity numeraire and the connection to the swap market model (SMM). For a fixed swap with start date T_0 , payment dates $T_1 < \dots < T_n$, and accruals (α_j) , define the swap annuity

$$A(t) := \sum_{j=1}^n \alpha_j B(t, T_j).$$

The associated annuity measure $\mathbb{Q}^A$ is the numeraire measure induced by $A(\cdot)$ via (24), applied on the pricing interval $[t, T_0]$ with exercise horizon $T = T_0$. Under $\mathbb{Q}^A$, the fair swap rate

$$R^{\text{swap}}(t) := \frac{B(t, T_0) - B(t, T_n)}{A(t)}$$

is modeled directly in the SMM, often as lognormal. This is consistent with the same engine structure: when a payoff is naturally expressed in annuity units, for example as $A(T_0) h(R^{\text{swap}}(T_0))$, digital and CDF blocks become probabilities under $\mathbb{Q}^A$. SMM therefore appears as another natural numeraire instantiation within the NDSE framework, complementary to bond forward measures.

5 First-order sensitivity template continuing the DSE framework

The *Distributional Sensitivity Engine* (DSE) first-order engine is obtained by viewing (w_1, w_2) as functions of an $\mathcal{F}_t$ -measurable boundary and a small set of “distributional” ingredients. The stochastic-discounting extension changes only the identity of the second measure, from $\mathbb{Q}$ to $\mathbb{Q}^T$, and the boundary variable, from m to $\tilde{m}$.

Let η denote a generic scalar distributional parameter entering the conditional law of $\tilde{E}_{t,T}$ under the relevant measures. Define the call-oriented conditional CDF blocks

$$\Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbf{M}}(m; \eta) := \mathbb{P}_t^{\mathbf{M}}(\tilde{E}_{t,T} < m), \quad \mathbf{M} \in \{\mathbb{Q}^+, \mathbb{Q}^T\}, \quad (33)$$

with corresponding conditional densities $\phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^+}$ and $\phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^T}$. Continuity of the conditional law implies $\mathbb{P}_t(\tilde{E}_{t,T} = \tilde{m}) = 0$. The signed digital weights appearing in (23) admit the equivalent representation

$$w_1(\tilde{m}; \eta) = \Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^+}(\tilde{m}; \eta) - \frac{1-D}{2}, \quad w_2(\tilde{m}; \eta) = \Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^T}(\tilde{m}; \eta) - \frac{1-D}{2}, \quad (34)$$

so the constant shift only affects the price level through put–call parity and disappears in derivatives. Under the same conditional regularity assumptions as in DSE, namely, the existence and differentiability of conditional densities, the chain rule yields for any scalar parameter x :

$$\frac{\partial V_t}{\partial x} = w_1 \frac{\partial a_1}{\partial x} + w_2 \frac{\partial a_2}{\partial x} + w_3 \frac{\partial \eta}{\partial x} + w_4 \frac{\partial \tilde{m}}{\partial x}, \quad (35)$$

where the two additional weights are

$$w_3 := a_1 \partial_\eta \Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^+}(z; \eta)|_{z=\tilde{m}} + a_2 \partial_\eta \Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^T}(z; \eta)|_{z=\tilde{m}}, \quad (36)$$

$$w_4 := a_1 \phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^+}(\tilde{m}; \eta) + a_2 \phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^T}(\tilde{m}; \eta). \quad (37)$$

This is the stochastic-discounting analogue of the four-input engine in DSE. Mixed partials follow the same organizational logic as in DSE. They can be arranged as a dot product between common weights and second derivatives of primitives together with an explicit correction channel that collects weight-motion products.

6 Minimal Dirac blocks for differentiating digital weights in Monte Carlo backends

The representation (35) reduces sensitivity computation to evaluating conditional CDF and PDF blocks, including their parameter derivatives, at the boundary $\tilde{m}$. This section records the core Dirac-delta identities needed to express and differentiate the digital and CDF blocks in simulation without finite differences [11, 12]. The focus is on kernel mollification and likelihood-ratio reweighting, which are sufficient for integration into the engine. Further distributional analysis is deferred.

6.1 CDF and PDF as expectations of indicators and deltas

For any real-valued random variable Y with density, the distribution function and density satisfy

$$\Phi_Y(m) = \mathbb{E}[\mathbf{1}_{\{Y < m\}}], \quad \phi_Y(m) = \frac{d}{dm} \Phi_Y(m) = \mathbb{E}[\delta(Y - m)], \quad (38)$$

where δ is the Dirac delta in the distributional sense. Applied to the conditional laws needed in (36)–(37),

$$\phi_{\tilde{E}|_{\mathcal{F}_t}}^{\mathbf{M}}(\tilde{m}; \eta) = \mathbb{E}_t^{\mathbf{M}}[\delta(\tilde{E}_{t,T} - \tilde{m})], \quad \mathbf{M} \in \{\mathbb{Q}^+, \mathbb{Q}^T\}. \quad (39)$$

6.2 Parameter derivatives of CDF blocks via δ

Suppose $\tilde{E}_{t,T} = \tilde{E}_{t,T}(\theta)$ depends smoothly on a scalar model parameter θ (e.g., a volatility parameter, correlation, or a rate-model parameter). Formally differentiating the indicator while holding the threshold argument fixed gives the distributional identity

$$\frac{\partial}{\partial \theta} \mathbf{1}_{\{\tilde{E}_{t,T}(\theta) < \tilde{m}\}} = -\delta(\tilde{E}_{t,T}(\theta) - \tilde{m}) \frac{\partial \tilde{E}_{t,T}}{\partial \theta}. \quad (40)$$

Conditional expectations under a measure $\mathbf{M}$ yield the identity below. Differentiation under the expectation follows under the regularity conditions in Appendix A.

$$\partial_\theta \Phi_{\tilde{E}|_{\mathcal{F}_t}}^{\mathbf{M}}(\tilde{m}; \eta) = -\mathbb{E}_t^{\mathbf{M}}[\delta(\tilde{E}_{t,T} - \tilde{m}) \partial_\theta \tilde{E}_{t,T}]. \quad (41)$$

This identity is relevant when the engine requires $\partial_\eta \Phi$ blocks but the density itself is available only through simulation.

6.3 Computing blocks under $\mathbb{Q}$ via likelihood ratios

In Monte Carlo, it is often simplest to simulate paths under the baseline measure $\mathbb{Q}$ and reweight. Define the likelihood ratios, namely the Radon–Nikodym derivatives, [13]

$$L^+ := \frac{d\mathbb{Q}^+}{d\mathbb{Q}} \Big|_{\mathcal{F}_T} = e^{-E_{t,T}}, \quad L^T := \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \Big|_{\mathcal{F}_T} = \frac{\text{DF}_{t,T}}{B(t,T)}. \quad (42)$$

Here $L^T = L_T^T$ is the terminal value of the density process L_u^T in (7); similarly $L^+ = L_T^+$ is the terminal value of the density process defining $\mathbb{Q}^+$ in (19).

In the HJM setting, (8) gives the explicit form

$$L^T = \mathcal{E} \left(- \int_t^T \Sigma(u, T)^\top dW_u^r \right) = \exp \left(- \int_t^T \Sigma(u, T)^\top dW_u^r - \frac{1}{2} \int_t^T \|\Sigma(u, T)\|^2 du \right). \quad (43)$$

This representation is well suited to Monte Carlo: one may simulate the term-structure factors under $\mathbb{Q}$, or directly the forward-rate curve on a maturity grid, accumulate the Brownian integral of $\Sigma(\cdot, T)$, and evaluate L^T without separately estimating $B(t, T)$ on each path.

Then the CDF and PDF blocks can be written as $\mathbb{Q}$ -expectations:

$$\Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^+}(\tilde{m}; \eta) = \mathbb{E}_t^{\mathbb{Q}}[L^+ \mathbf{1}_{\{\tilde{E}_{t,T} < \tilde{m}\}}], \quad (44)$$

$$\Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^T}(\tilde{m}; \eta) = \mathbb{E}_t^{\mathbb{Q}}[L^T \mathbf{1}_{\{\tilde{E}_{t,T} < \tilde{m}\}}], \quad (45)$$

$$\phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^+}(\tilde{m}; \eta) = \mathbb{E}_t^{\mathbb{Q}}[L^+ \delta(\tilde{E}_{t,T} - \tilde{m})], \quad (46)$$

$$\phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^T}(\tilde{m}; \eta) = \mathbb{E}_t^{\mathbb{Q}}[L^T \delta(\tilde{E}_{t,T} - \tilde{m})]. \quad (47)$$

For the *signed* digital weights that enter the price level (Section 4), one may equivalently estimate

$$w_1 = D \mathbb{E}_t^{\mathbb{Q}}[L^+ \mathbf{1}_{\{D\tilde{E}_{t,T} < D\tilde{m}\}}], \quad w_2 = D \mathbb{E}_t^{\mathbb{Q}}[L^T \mathbf{1}_{\{D\tilde{E}_{t,T} < D\tilde{m}\}}],$$

which is consistent with (20)–(22). By contrast, (44)–(45) are the call-oriented CDF blocks used in (34), and hence in the derivative weights (36)–(37). Similarly, for a parameter θ entering $\tilde{E}_{t,T}$ while the likelihood ratios are held fixed,

$$\begin{aligned} \partial_\theta \Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^+}(\tilde{m}; \eta) &= -\mathbb{E}_t^{\mathbb{Q}}[L^+ \delta(\tilde{E}_{t,T} - \tilde{m}) \partial_\theta \tilde{E}_{t,T}], \\ \partial_\theta \Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^T}(\tilde{m}; \eta) &= -\mathbb{E}_t^{\mathbb{Q}}[L^T \delta(\tilde{E}_{t,T} - \tilde{m}) \partial_\theta \tilde{E}_{t,T}]. \end{aligned} \quad (48)$$

If θ also enters the likelihood ratios L^+ or L^T , the corresponding derivative terms must be added separately. The formulas in (48) isolate only the derivative channel coming from the boundary variable $\tilde{E}_{t,T}$.

6.4 Practical mollifiers for δ

In computation, replace δ by a smooth approximation (mollifier) δ_ε , for example:

$$\delta_\varepsilon^{\text{Gauss}}(x) := \frac{1}{\sqrt{2\pi}\varepsilon} \exp\left(-\frac{x^2}{2\varepsilon^2}\right), \quad (49)$$

$$\delta_\varepsilon^{\text{box}}(x) := \frac{1}{2\varepsilon} \mathbf{1}_{\{|x| \leq \varepsilon\}}. \quad (50)$$

Then, for N Monte Carlo paths $\{\tilde{E}^{(i)}, L^{(i)}\}_{i=1}^N$, a plug-in estimator for the PDF block is

$$\hat{\phi}^{\mathbf{M}}(\tilde{m}) := \frac{1}{N} \sum_{i=1}^N L^{\mathbf{M},(i)} \delta_\varepsilon(\tilde{E}^{(i)} - \tilde{m}), \quad \mathbf{M} \in \{\mathbb{Q}^+, \mathbb{Q}^T\}. \quad (51)$$

In practice, the choice of ε can follow standard kernel-density heuristics. Out-of-sample error minimization for ϕ can also guide the choice. The engine framework does not impose a specific rule. Stability of the resulting blocks is required.

6.5 Plugging the blocks into the sensitivity engine

The first-order engine (35) requires (w_1, w_2, w_3, w_4) . Given Monte Carlo estimators for CDF and PDF blocks under $\mathbb{Q}^+$ and $\mathbb{Q}^T$ ((44)–(47)) and, if needed, parameter-derivative blocks ((48)), one obtains an estimator for $\partial_x V_t$ by evaluating

$$\hat{V}_{t,x} := \hat{w}_1 a_{1,x} + \hat{w}_2 a_{2,x} + \hat{w}_3 \eta_x + \hat{w}_4 \tilde{m}_x. \quad (52)$$

Here $a_{1,x}$, $a_{2,x}$, $\tilde{m}_x$ are deterministic or $\mathcal{F}_t$ -measurable derivatives, while $\hat{w}_k$ are computed from the simulated distributional blocks. This separation mirrors DSE . Once the distributional blocks are available, the sensitivity logic is unchanged.

Remark 6.1. Scope. The Dirac identities in this section are used only to obtain simulation-friendly derivatives of digital and CDF blocks. Connections to state-price densities and to more elaborate distributional estimators are left to subsequent work. Appendix G records the basic identities.

7 Conclusion

The paper presents a numeraire-based distributional sensitivity engine for European options under stochastic interest rates. The framework extends *Distributional Sensitivity Engine* (DSE) from deterministic discounting to stochastic discounting while retaining the same basic sensitivity-engine organization. The analysis is carried out for European options, and the main text assumes a deterministic dividend yield. The key step normalizes the stochastic discount factor by the bond price $B(t, T)$. This normalization yields an $\mathcal{F}_t$ -measurable exercise boundary for a modified Doleans–Dade log variable $\tilde{E}_{t,T}$. The spot leg remains under the Doleans tilt measure $\mathbb{Q}^+$, while the strike leg is represented through a digital weight under the T -forward measure, or more generally under a suitable numeraire measure. The resulting two-weight representation retains the four-input first-derivative template of DSE and remains compatible with the same mixed-partial correction organization.

The numerical implementation requires CDF and PDF evaluations together with parameter derivatives of digital weights. Dirac-delta representations yield simulation estimators for these blocks. A mollified delta yields practical estimators for PDF and derivative blocks with standard bias and variance behavior. Taken together, these ingredients yield a practical Monte Carlo implementation layer for sensitivity estimation in stochastic-rate settings.

A Python implementation corresponding to the present framework is available in GitHub: https://github.com/DavidHShen/Numeraire-Based_Distributional_Sensitivity_Engine.

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Appendix

A Regularity remarks for δ -based differentiation

The identities (38)–(41) are distributional. To justify exchanging differentiation and conditional expectation, sufficient conditions include: (i) existence of a conditional density for $\tilde{E}_{t,T}$ under the measure of interest; (ii) integrability of the likelihood ratio L used to rewrite expectations under $\mathbb{Q}$; and (iii) dominated-convergence-type bounds for the mollified estimators. In practice, the mollifier approach replaces the singular object δ by a smooth kernel δ_ε so that differentiation under the expectation becomes a standard calculus step for each fixed ε , and the $\varepsilon \downarrow 0$ limit is treated as a bias–variance trade-off rather than a strict limit.

B HJM drift restriction and bond dynamics

This appendix derives the HJM drift restriction (28) and the bond dynamics (4) used in the main text. Fix a maturity $U > u$ and define the log-bond price

$$\ell(u; U) := \ln B(u, U) = - \int_u^U f(u, s) ds. \quad (53)$$

Because the lower integration limit depends on u , the stochastic Leibniz rule gives

$$d\left(\int_u^U f(u, s) ds\right) = \int_u^U df(u, s) ds - f(u, u) du. \quad (54)$$

Therefore,

$$d\ell(u; U) = - \int_u^U df(u, s) ds + f(u, u) du. \quad (55)$$

Substituting the HJM forward dynamics (27) yields

$$\begin{aligned} d\ell(u; U) &= \left(r_u - \int_u^U \alpha(u, s) ds\right) du - \left(\int_u^U \sigma_f(u, s) ds\right)^\top dW_u^r \\ &= \left(r_u - \int_u^U \alpha(u, s) ds\right) du - \Sigma(u, U)^\top dW_u^r. \end{aligned} \quad (56)$$

By Itô's formula for $B(u, U) = e^{\ell(u; U)}$,

$$\frac{dB(u, U)}{B(u, U)} = d\ell(u; U) + \frac{1}{2} d\langle \ell(\cdot; U) \rangle_u = \left(r_u - \int_u^U \alpha(u, s) ds + \frac{1}{2} \|\Sigma(u, U)\|^2\right) du - \Sigma(u, U)^\top dW_u^r. \quad (57)$$

Under the money-market numeraire measure $\mathbb{Q}$, discounted bond prices must be local martingales, so the drift in (57) must equal r_u . Hence

$$\int_u^U \alpha(u, s) ds = \frac{1}{2} \|\Sigma(u, U)\|^2. \quad (58)$$

Differentiating (58) with respect to U gives the pointwise HJM drift restriction

$$\alpha(u, U) = \Sigma(u, U)^\top \sigma_f(u, U) = \sigma_f(u, U)^\top \int_u^U \sigma_f(u, s) ds, \quad (59)$$

which is (28). Substituting (58) back into (57) yields the bond dynamics (4):

$$\frac{dB(u, U)}{B(u, U)} = r_u du - \Sigma(u, U)^\top dW_u^r.$$

C T -forward measure likelihood ratio

Define the process

$$L_u^T := \frac{\text{DF}_{t,u} B(u, T)}{B(t, T)}, \quad u \in [t, T]. \quad (60)$$

The definition $\text{DF}_{t,u} = \exp(-\int_t^u r_s ds)$ implies $d\text{DF}_{t,u}/\text{DF}_{t,u} = -r_u du$. Combining this with the bond SDE (4) and applying the product rule,

$$\begin{aligned} \frac{dL_u^T}{L_u^T} &= \frac{d\text{DF}_{t,u}}{\text{DF}_{t,u}} + \frac{dB(u,T)}{B(u,T)} + d\langle \ln \text{DF}_{t,\cdot}, \ln B(\cdot, T) \rangle_u \\ &= (-r_u du) + (r_u du - \Sigma(u, T)^\top dW_u^r) + 0 = -\Sigma(u, T)^\top dW_u^r. \end{aligned} \quad (61)$$

Thus the process $(L_u^T)_{u \in [t, T]}$ is a strictly positive local martingale, and under standard integrability conditions (e.g. Novikov) it is a true martingale. Evaluating (60) at $u = T$ gives

$$L_T^T = \frac{\text{DF}_{t,T}}{B(t, T)}.$$

Solving (61) yields the exponential-martingale representation (8):

$$\frac{\text{DF}_{t,T}}{B(t, T)} = \mathcal{E}\left(-\int_t^T \Sigma(u, T)^\top dW_u^r\right) = \exp\left(-\int_t^T \Sigma(u, T)^\top dW_u^r - \frac{1}{2} \int_t^T \|\Sigma(u, T)\|^2 du\right).$$

Moreover, by Girsanov, under $\mathbb{Q}^T$ the rate Brownian motion is shifted as

$$dW_u^{r, \mathbb{Q}^T} = dW_u^r + \Sigma(u, T) du, \quad (62)$$

which is the standard forward-measure drift adjustment under the bond numeraire when $\frac{dB(u, T)}{B(u, T)} = r_u du - \Sigma(u, T)^\top dW_u^r$ holds under $\mathbb{Q}$.

D Equity stochastic exponential and event rewriting

Starting from the equity SDE (9), Itô's formula gives

$$d \ln S_u = \left(r_u - q(u) - \frac{1}{2} \|\sigma_S(u, \cdot)\|^2\right) du + \sigma_S(u, \cdot)^\top dW_u^S. \quad (63)$$

Integrating (63) from t to T yields

$$S_T = S_t \exp\left(\int_t^T (r_u - q(u)) du\right) \exp\left(\int_t^T \sigma_S(u, \cdot)^\top dW_u^S - \frac{1}{2} \int_t^T \|\sigma_S(u, \cdot)\|^2 du\right). \quad (64)$$

The last exponential is precisely the Doleans–Dade exponential $\mathcal{E}\left(\int_t^T \sigma_S(u, \cdot)^\top dW_u^S\right)$, which yields (10)–(11). Defining

$$E_{t,T} := -\ln \mathcal{E}\left(\int_t^T \sigma_S(u, \cdot)^\top dW_u^S\right), \quad (65)$$

the definitions $\text{DF}_{t,T} = \exp(-\int_t^T r_u du)$ and $Q_{t,T} = \exp(\int_t^T q(u) du)$ imply that the identity (11) is equivalent to

$$\ln S_T = \ln S_t - \ln \text{DF}_{t,T} - \ln Q_{t,T} - E_{t,T}. \quad (66)$$

Adding and subtracting $\ln B(t, T)$ motivates the numeraire-adjusted variable

$$\tilde{E}_{t,T} := E_{t,T} + \ln\left(\frac{\text{DF}_{t,T}}{B(t, T)}\right), \quad \tilde{m} := \ln\left(\frac{S_t}{K Q_{t,T} B(t, T)}\right),$$

which recover (13). A direct rearrangement gives the useful identity

$$\tilde{E}_{t,T} - \tilde{m} = \ln\left(\frac{K}{S_T}\right), \quad (67)$$

so that the in-the-money event is equivalent to (16):

$$\mathbf{1}_{\{DS_T > DK\}} = \mathbf{1}_{\{D\tilde{E}_{t,T} < D\tilde{m}\}}.$$

E Spot-leg tilt and strike-leg numeraire change

Spot leg. Define the density process

$$L_u^+ := \mathcal{E}\left(\int_t^u \sigma_S(s, \cdot)^\top dW_s^S\right), \quad u \in [t, T]. \quad (68)$$

Under a standard exponential-martingale condition (e.g. Novikov), the process $(L_u^+)_{u \in [t, T]}$ is a martingale with $L_t^+ = 1$. Define $\mathbb{Q}^+$ on $\mathcal{F}_T$ by $d\mathbb{Q}^+/d\mathbb{Q}|_{\mathcal{F}_T} = L_T^+$, which is (19). Then for any $\mathcal{F}_T$ -measurable random variable Y ,

$$\mathbb{E}_t^{\mathbb{Q}}[L_T^+ Y] = \mathbb{E}_t^{\mathbb{Q}^+}[Y]. \quad (69)$$

Taking $Y = \mathbf{1}_{\{D\tilde{E}_{t,T} < D\tilde{m}\}}$ and using $D\mathbb{F}_{t,T}S_T = (S_t/Q_{t,T})L_T^+$ gives the spot-leg identity (20).

Strike leg. The strike leg uses the standard change-of-numeraire identity. With the T -bond numeraire, for any $\mathcal{F}_T$ -measurable X_T ,

$$\mathbb{E}_t^{\mathbb{Q}}[D\mathbb{F}_{t,T}X_T] = B(t, T) \mathbb{E}_t^{\mathbb{Q}^T}[X_T],$$

which is (21). Choosing $X_T = \mathbf{1}_{\{D\tilde{E}_{t,T} < D\tilde{m}\}}$ yields (22).

F Derivation of the four-input first-order engine

Write the two-weight form (23) as $V_t = a_1 w_1(\tilde{m}; \eta) + a_2 w_2(\tilde{m}; \eta)$, where

$$w_1(\tilde{m}; \eta) = \Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^+}(\tilde{m}; \eta) - \frac{1-D}{2}, \quad w_2(\tilde{m}; \eta) = \Phi_{\tilde{E}|\mathcal{F}_t}^{\mathbb{Q}^T}(\tilde{m}; \eta) - \frac{1-D}{2}.$$

The constant shift $(1-D)/2$ is independent of x and drops out of all derivative terms. Let x be a scalar parameter that may enter (a_1, a_2) directly and may also enter the conditional law of $\tilde{E}_{t,T}$ through a distributional parameter η and through the boundary $\tilde{m}$. Differentiating and applying the chain rule gives

$$\begin{aligned} \partial_x V_t &= w_1 \partial_x a_1 + w_2 \partial_x a_2 + a_1 \left(\partial_\eta \Phi^{\mathbb{Q}^+}(\tilde{m}; \eta) \partial_x \eta + \phi^{\mathbb{Q}^+}(\tilde{m}; \eta) \partial_x \tilde{m} \right) \\ &\quad + a_2 \left(\partial_\eta \Phi^{\mathbb{Q}^T}(\tilde{m}; \eta) \partial_x \eta + \phi^{\mathbb{Q}^T}(\tilde{m}; \eta) \partial_x \tilde{m} \right), \end{aligned} \quad (70)$$

which yields (35) with

$$w_3 = a_1 \partial_\eta \Phi^{\mathbb{Q}^+}(\tilde{m}; \eta) + a_2 \partial_\eta \Phi^{\mathbb{Q}^T}(\tilde{m}; \eta), \quad w_4 = a_1 \phi^{\mathbb{Q}^+}(\tilde{m}; \eta) + a_2 \phi^{\mathbb{Q}^T}(\tilde{m}; \eta).$$

The mixed-partial organization stated in the main text follows by applying ∂_y to $\partial_x V_t$ and collecting terms into (i) second derivatives of the primitives and (ii) products of first derivatives of weights and primitives, as in *Distributional Sensitivity Engine* (DSE) .

G Dirac-delta identities, Jacobians, and Breeden–Litzenberger

CDF and PDF identities. For a real random variable Y with density, the distributional derivative satisfies

$$\frac{d}{dm} \mathbf{1}_{\{Y < m\}} = \delta(Y - m)$$

in the sense of test functions. Taking expectations yields (38): $\phi_Y(m) = \mathbb{E}[\delta(Y - m)]$.

Parameter derivatives. For a smooth parametrization $Y = Y(\theta)$, differentiation of $\mathbf{1}_{\{Y(\theta) < m\}}$ gives the distributional chain rule

$$\partial_\theta \mathbf{1}_{\{Y(\theta) < m\}} = -\delta(Y(\theta) - m) \partial_\theta Y(\theta),$$

which is (40). Taking conditional expectations yields (41).

Jacobian relation for the exercise boundary. The identity (67) implies $\tilde{E}_{t,T} - \tilde{m} = \ln(K/S_T)$. Since $g(s) := \ln(K/s)$ has derivative $g'(s) = -1/s$, the change-of-variables rule for the Dirac delta gives

$$\delta(\tilde{E}_{t,T} - \tilde{m}) = \delta(\ln(K/S_T)) = \frac{\delta(S_T - K)}{|g'(K)|} = K \delta(S_T - K), \quad (71)$$

so $\delta(S_T - K) = \frac{1}{K} \delta(\tilde{E}_{t,T} - \tilde{m})$.

Breeden–Litzenberger. For a call with strike K , $V_t^{\text{call}}(K) = \mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T}(S_T - K)^+]$ [3]. Under conditions allowing differentiation under the expectation,

$$\frac{\partial}{\partial K} V_t^{\text{call}}(K) = -\mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T} \mathbf{1}_{\{S_T > K\}}],$$

and differentiating once more yields

$$\frac{\partial^2}{\partial K^2} V_t^{\text{call}}(K) = \mathbb{E}_t^{\mathbb{Q}}[\text{DF}_{t,T} \delta(S_T - K)]. \quad (72)$$

Combining (72) with (71) links the state-price density at the strike to the δ -based PDF blocks in the engine.

The one-dimensional change-of-variables identity (71) is a special case of a general “surface Jacobian” rule. Let Z be an $\mathbb{R}^d$ -valued random vector with density f_Z and let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be C^1 . Assume m is a regular value of g (that is, $\nabla g(z) \neq 0$ on the level set $\{z : g(z) = m\}$). Then for any integrable test function ψ ,

$$\mathbb{E}[\psi(Z) \delta(g(Z) - m)] = \int_{g(z)=m} \frac{\psi(z) f_Z(z)}{\|\nabla g(z)\|} dS(z), \quad (73)$$

where dS denotes the $(d-1)$ -dimensional surface measure on the level set. In particular,

$$\mathbb{E}[\delta(g(Z) - m)] = \int_{g(z)=m} \frac{f_Z(z)}{\|\nabla g(z)\|} dS(z), \quad (74)$$

which makes precise the geometric meaning of Dirac blocks as “mass on the boundary.”

Let K be a smooth probability density on $\mathbb{R}$ and define the mollifier $\delta_\varepsilon(x) := \varepsilon^{-1}K(x/\varepsilon)$. For fixed $\varepsilon > 0$,

$$\mathbb{E}[\psi(Z) \delta_\varepsilon(g(Z) - m)] = \int_{\mathbb{R}^d} \psi(z) f_Z(z) \delta_\varepsilon(g(z) - m) dz.$$

$$\int_{\mathbb{R}^d} h(z) dz = \int_{\mathbb{R}} \int_{g(z)=u} \frac{h(z)}{\|\nabla g(z)\|} dS(z) du.$$

Applying this with $h(z) := \psi(z) f_Z(z) \delta_\varepsilon(g(z) - m)$ gives

$$\int_{\mathbb{R}^d} \psi(z) f_Z(z) \delta_\varepsilon(g(z) - m) dz = \int_{\mathbb{R}} \delta_\varepsilon(u - m) \left(\int_{g(z)=u} \frac{\psi(z) f_Z(z)}{\|\nabla g(z)\|} dS(z) \right) du.$$

In one dimension with g monotone, the surface integral reduces to the usual Jacobian rule. This recovers (71).

G.1 Higher-order distributional blocks: δ' and density sensitivities

When differentiating densities or boundary blocks further, the derivative δ' appears. Two basic identities are:

$$\partial_m \mathbb{E}[\delta(X - m)] = -\mathbb{E}[\delta'(X - m)], \quad (75)$$

$$\partial_\theta \mathbb{E}[\delta(X(\theta) - m)] = -\mathbb{E}[\delta'(X(\theta) - m) \partial_\theta X(\theta)], \quad (76)$$

interpreted in the distributional sense (test-function integration by parts). In Monte Carlo, a natural implementation uses the derivative of a smooth kernel. If $\delta_\varepsilon(x) = \varepsilon^{-1}K(x/\varepsilon)$ with differentiable K , then

$$\delta'_\varepsilon(x) = \frac{d}{dx} \delta_\varepsilon(x) = \frac{1}{\varepsilon^2} K'\left(\frac{x}{\varepsilon}\right), \quad (77)$$

so that expectations involving δ' can be estimated by replacing δ' with δ'_ε . Because (77) introduces an additional factor ε^{-1} relative to δ_ε , variance typically increases; a corresponding bias-variance balance often scales as $\text{MSE} \approx O(\varepsilon^4) + O((N\varepsilon^3)^{-1})$. This implies $\varepsilon \asymp N^{-1/7}$ for δ' -type blocks under smoothness assumptions.

G.2 Alternatives and extensions: IBP, local time, and Fourier benchmarks

Dirac as a conditional-expectation operator. Whenever $f_X(m) = \mathbb{E}[\delta(X - m)] > 0$, one has the level-set conditioning identity

$$\mathbb{E}[g(Y) \mid X = m] = \frac{\mathbb{E}[g(Y) \delta(X - m)]}{\mathbb{E}[\delta(X - m)]},$$

which can be useful for “boundary diagnostics” (e.g. conditional factor moments on the exercise boundary).

Integration-by-parts estimators using Malliavin or score methods. A representative form, under Malliavin differentiability and non-degeneracy, expresses the density as [14]

$$f_X(m) = \mathbb{E}[\delta(X - m)] = \mathbb{E}[\mathbf{1}_{\{X > m\}} H],$$

for a suitable integration-by-parts weight H depending on the model and on X . This alternative is not pursued here.

Barrier and hitting events with local time. For pathwise boundaries (e.g. barriers), continuous-time limits involve local time. The occupation density formula implies that, for a semimartingale S ,

$$L_T^b = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \int_0^T \mathbf{1}_{\{|S_u - b| \leq \varepsilon\}} d\langle S \rangle_u,$$

so mollifier-based Dirac blocks can be interpreted as discrete approximations of boundary-contact and local-time contributions.

Fourier representation. Dirac admits the Fourier representation in the sense of tempered distributions

$$\delta(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{iux} du,$$

which yields the inversion formula

$$f_X(m) = \mathbb{E}[\delta(X - m)] = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ium} \varphi_X(u) du, \quad \varphi_X(u) := \mathbb{E}[e^{iuX}],$$

whenever the characteristic function φ_X is available and the inversion is well-defined. This provides semi-analytic benchmarks for validating mollifier-based Monte Carlo estimates of Dirac blocks.

Universal interface viewpoint. Across the extensions above, the basic NDSE “shell” remains unchanged: model dependence enters the Dirac layer only through (i) the distribution of the boundary variable X under the relevant measures and (ii) the likelihood ratios used to express those measures under a common simulation measure.

H Bias–variance and bandwidth scaling for mollified Dirac estimators

This appendix records a standard bias–variance calculation for the mollified Dirac estimators used in Section 6. Let Y be a real-valued random variable with conditional density $\phi_Y^{\mathbf{M}}$ under the measure $\mathbf{M}$ given $\mathcal{F}_t$, and consider the weighted PDF block

$$g_Y^{\mathbf{M}}(m) := \mathbb{E}_t^{\mathbf{M}}[L \delta(Y - m)],$$

where L is an integrable weight under the measure $\mathbf{M}$. Let K be a smooth probability density on $\mathbb{R}$ and define the mollifier $\delta_\varepsilon(x) := \varepsilon^{-1} K(x/\varepsilon)$. For i.i.d. samples $\{(Y^{(i)}, L^{(i)})\}_{i=1}^N$ under $\mathbf{M}$, the kernel estimator is

$$\hat{g}_Y^{\mathbf{M}}(m) := \frac{1}{N} \sum_{i=1}^N L^{(i)} \delta_\varepsilon(Y^{(i)} - m).$$

Assume $m \mapsto g_Y^{\mathbf{M}}(m)$ is twice continuously differentiable and that K is symmetric with finite second moment $\mu_2(K) := \int u^2 K(u) du$. Then a standard Taylor expansion yields the conditional bias

$$\mathbb{E}_t^{\mathbf{M}}[\hat{g}_Y^{\mathbf{M}}(m)] - g_Y^{\mathbf{M}}(m) = \frac{\varepsilon^2}{2} \mu_2(K) \partial_{mm} g_Y^{\mathbf{M}}(m) + O(\varepsilon^4).$$

Moreover, provided $\int K(u)^2 du < \infty$ and $\mathbb{E}_t^{\mathbf{M}}[L^2] < \infty$, the leading conditional variance scales as

$$\text{Var}_t^{\mathbf{M}}(\hat{g}_Y^{\mathbf{M}}(m)) = \frac{1}{N} \left(\mathbb{E}_t^{\mathbf{M}}[L^2 \delta_\varepsilon(Y - m)^2] - (\mathbb{E}_t^{\mathbf{M}}[L \delta_\varepsilon(Y - m)])^2 \right).$$

As $\varepsilon \downarrow 0$, one has $\delta_\varepsilon(x)^2 = \varepsilon^{-1} R(K) \delta(x) + o(\varepsilon^{-1})$ in the distributional sense, where $R(K) := \int K(u)^2 du$. Hence

$$\text{Var}_t^{\mathbf{M}}(\hat{g}_Y^{\mathbf{M}}(m)) = \frac{1}{N\varepsilon} R(K) \mathbb{E}_t^{\mathbf{M}}[L^2 \delta(Y - m)] + o\left(\frac{1}{N\varepsilon}\right).$$

If a conditional density exists under $\mathbf{M}$, then $\mathbb{E}_t^{\mathbf{M}}[L^2 \delta(Y - m)] = \phi_Y^{\mathbf{M}}(m) \mathbb{E}_t^{\mathbf{M}}[L^2 \mid Y = m]$. Consequently, the conditional mean-squared error admits the scaling

$$\text{MSE}_t(\hat{g}_Y^{\mathbf{M}}(m)) = O(\varepsilon^4) + O\left(\frac{1}{N\varepsilon}\right),$$

and the balancing choice satisfies $\varepsilon_\star = O(N^{-1/5})$ in the usual smooth-density regime. This heuristic guides bandwidth selection for the PDF and Dirac-derivative blocks in Section 6.